

Bypass SSL Pinning – Complete Guide

SSL Pinning Kya Hai?

SSL pinning ek security mechanism hai jo mobile apps ko Man-in-the-Middle (MITM) attacks se bachati hai. Ismein app sirf specific certificates ya public keys ko trust karti hai, na ke system ke poore certificate store ko . Jab aap Burp Suite jaise proxy use karte hain, toh app connection reject kar deti hai agar proxy ka certificate trusted nahi hai.

Quick Method 1: Objection (Easiest)

Agar Frida server setup hai, toh Objection sab se easy tareeqa hai:

```
objection -g com.example.app explore  
android sslpinning disable
```

Yeh command automatically popular SSL pinning methods ko disable kar deti hai . Objection ke andar direct android sslpinning disable command use karein .

Quick Method 2: Universal Frida Scripts

Method A: frida-multiple-unpinning

```
frida -U -f com.example.app -l frida-multiple-unpinning.js --no-pause
```

Yeh script multiple SSL implementations (TrustManager, OkHttp, WebView, etc.) ko handle karti hai .

Method B: Universal Bypass Script

```
frida -U -l root_bypass.js -f com.target.application
```

Advanced Method: Flutter Apps Ke Liye

Flutter apps traditional methods se bypass nahi hoti kyunke Flutter apna custom SSL implementation (BoringSSL) use karta hai, jo system proxy settings aur certificates ko ignore karta hai .

Process:

1. **Proxy Setup:** Burp Suite configure karein aur Invisible Proxying enable karein
2. **Traffic Redirect:** ProxyDroid (root required) ya iptables use karein
3. **libflutter.so Analyze Karein:**
 - a. APKTool se APK decompile karein

b.Ghidra se libflutter.so mein
ssl_session_verify_cert_chain function locate
karein

4. Frida Script: Function offset calculate karein aur return
value manipulate karein

Method 3: Complete Interception Suite

HTTP Toolkit ka interception script saari cheezein handle
karta hai: proxy redirection, system certificate injection,
unpinning, root detection bypass, aur HTTP/3 blocking .

```
frida -U \  
  -l ./config.js \  
  -l ./native-connect-hook.js \  
  -l ./native-tls-hook.js \  
  -l ./android/android-proxy-override.js \  
  -l ./android/android-system-certificate-  
injection.js \  
  -l ./android/android-certificate-unpinning.js  
\  
  -l ./android/android-certificate-unpinning-  
fallback.js \  
  -l ./android/android-disable-root-detection.js  
\  
  -f $PACKAGE_ID
```

Setup Requirements

1. Frida Tools Install Karein:

```
pip install frida-tools
```

2. Frida Server Deploy Karein:

```
adb push frida-server /data/local/tmp/  
adb shell "chmod 755 /data/local/tmp/frida-server"  
adb shell "/data/local/tmp/frida-server &"
```

3. Burp Certificate Setup: ```bash

CERTIFICATE CONVERT KAREIN

```
openssl x509 -in certificate.pem -outform DER -out burp.crt
```

PHONE PAR PUSH KAREIN

```
adb push burp.crt /data/local/tmp/  
```
```

## Important Notes

- **Root Required:** Most methods require root access aur Frida server root permissions ke saath chalna chahiye
- **Version Matching:** PC aur phone par Frida versions match honi chahiye

- **Flutter Apps:** Traditional android sslpinning disable Flutter apps par kaam nahi karta
- **Detection:** Advanced security solutions (Appdome, etc.) Objection attach hone ko detect kar sakti hain

 **Educational Purpose Only:** SSL pinning bypass sirf authorized testing environments aur apne own apps mein karein. Kisi doosre ke app mein bina permission ke MITM testing karna applicable cyber laws ke under punishable hai.

---

MADE BY:  
SYED GHOU

---